HRMS

05 保險管理 — 系統設計書

目錄

HR05 INSURANCE Service — System Design

系統名稱	版本
人力資源暨專案管理系統	**1.0**
日期	作者
2026-03-06	**HRMS 開發團隊**

保險管理服務系統設計書

版本: 1.0
日期: 2025-12-07
Domain代號: 05 (INS)
目標: 提供工程師完整的系統實作規格，供PM建立工項清單

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10. 工項清單摘要

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1. 服務概述

1.1 服務定位

保險管理服務負責勞健保加退保自動化、投保級距管理、保險費用計算及政府申報檔案產生。本服務須確保符合台灣勞保、健保、勞退法規要求。

1.2 核心功能

- ✅ **多投保單位管理:** 母子公司分別投保
- ✅ **勞健保加退保自動化:** 訂閱員工事件自動處理
- ✅ **投保級距管理:** 自動對應、調整建議
- ✅ **費用計算:** 勞保、健保、勞退完整計算
- ✅ **二代健保補充保費:** 獎金超額2.11%計算

- ✅ **政府申報檔案:** 產生勞保局、健保局規範格式
- ✅ **團體保險方案管理:** 團體壽險/傷害險/醫療險方案 CRUD、職等對應、費用拆分
- ✅ **加退保生效日期規則:** 勞保=到職日、健保=到職日（月底離職例外）、團保=依合約

1.3 技術架構

- **前端:** ReactJS + Redux + Ant Design
- **後端:** Spring Boot 3.1.x + MyBatis
- **資料庫:** PostgreSQL 15.x
- **事件匯流排:** Kafka

1.4 服務邊界

屬於本服務	不屬於本服務
保險資料管理	員工資料 (Organization Service)
加退保處理	實際繳費 (Payroll提供數據)
費用計算	
申報檔案產生	

1.5 費率常數 (2025年)

項目	費率	個人負擔	雇主負擔	政府負擔
勞保	11.5%	20%	70%	10%
健保	5.17%	30%	60%	10%
勞退	6%	-	100%	-
補充保費	2.11%	100%	-	-

> *[!IMPORTANT] **法規參數可配置** 上述費率會依年度調整 (每年勞動部/衛福部公告)。系統須支援:*
>
> - *費率依 `effective_date` 生效日配置*
> - *級距表依年度版本管理*
> - *詳見: 法規參數管理與異動稽核設計規格書*
> - *詳見: 2025年度稅務保險級距表*
>
> ***生效日邏輯:***
>
> - *2026/1/1生效 = 適用2026年1月薪資 (2月發放)*

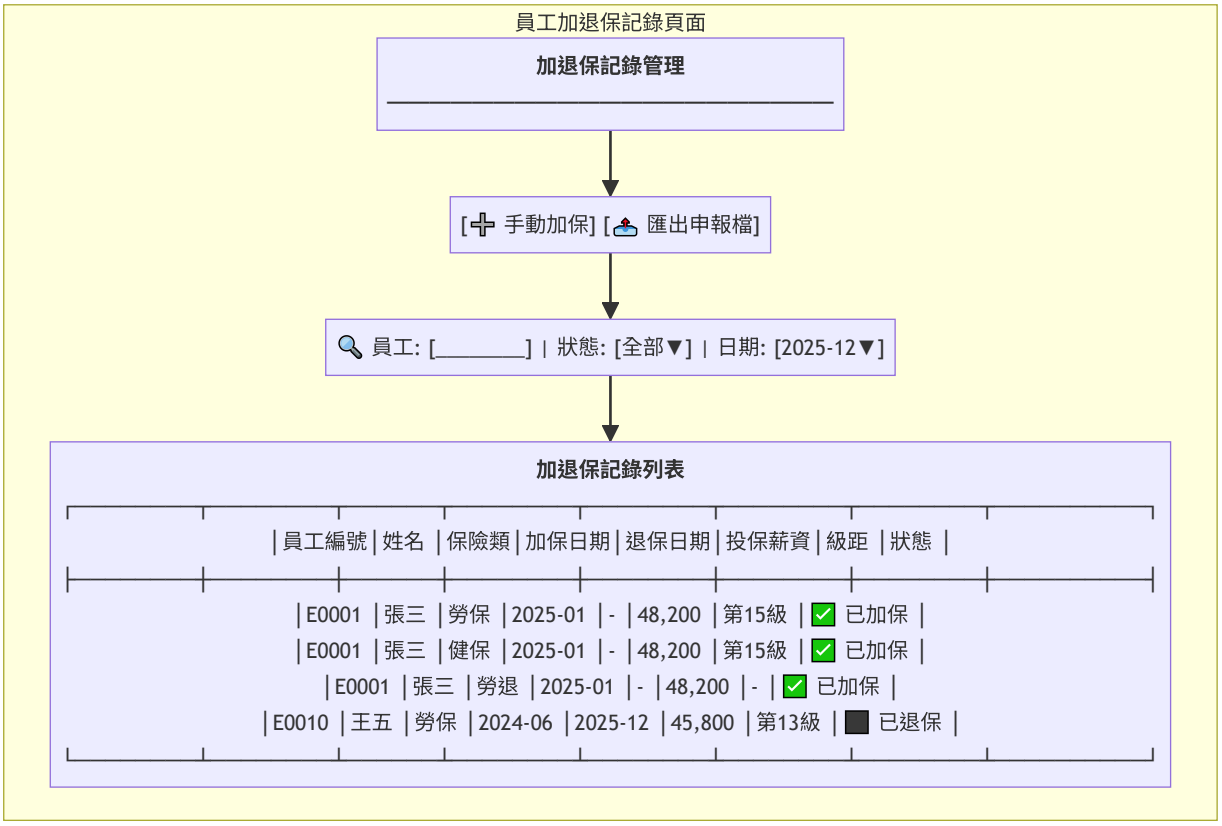
2. UI設計

2.1 頁面清單

頁面代碼	頁面名稱	路由	權限要求
`HR05-P01`	投保單位管理頁面	`/admin/insurance/units`	insurance:unit:manage
`HR05-P02`	員工加退保記錄頁面	`/admin/insurance/enrollments`	insurance:enrollment:read
`HR05-P03`	投保級距管理頁面	`/admin/insurance/levels`	insurance:level:manage
`HR05-P04`	保險費用試算頁面	`/admin/insurance/calculator`	insurance:calculate
`HR05-P05`	補充保費管理頁面	`/admin/insurance/supplementary`	insurance:supplementary:manage
`HR05-P06`	申報檔案產生頁面	`/admin/insurance/reports`	insurance:report:export
`HR05-P07`	我的保險資訊頁面 (ESS)	`/profile/insurance`	-
`HR05-M01`	手動加保對話框	(Modal)	insurance:enrollment:manage
`HR05-M02`	投保級距調整對話框	(Modal)	insurance:enrollment:manage

2.2 UI線稿 (Mermaid)

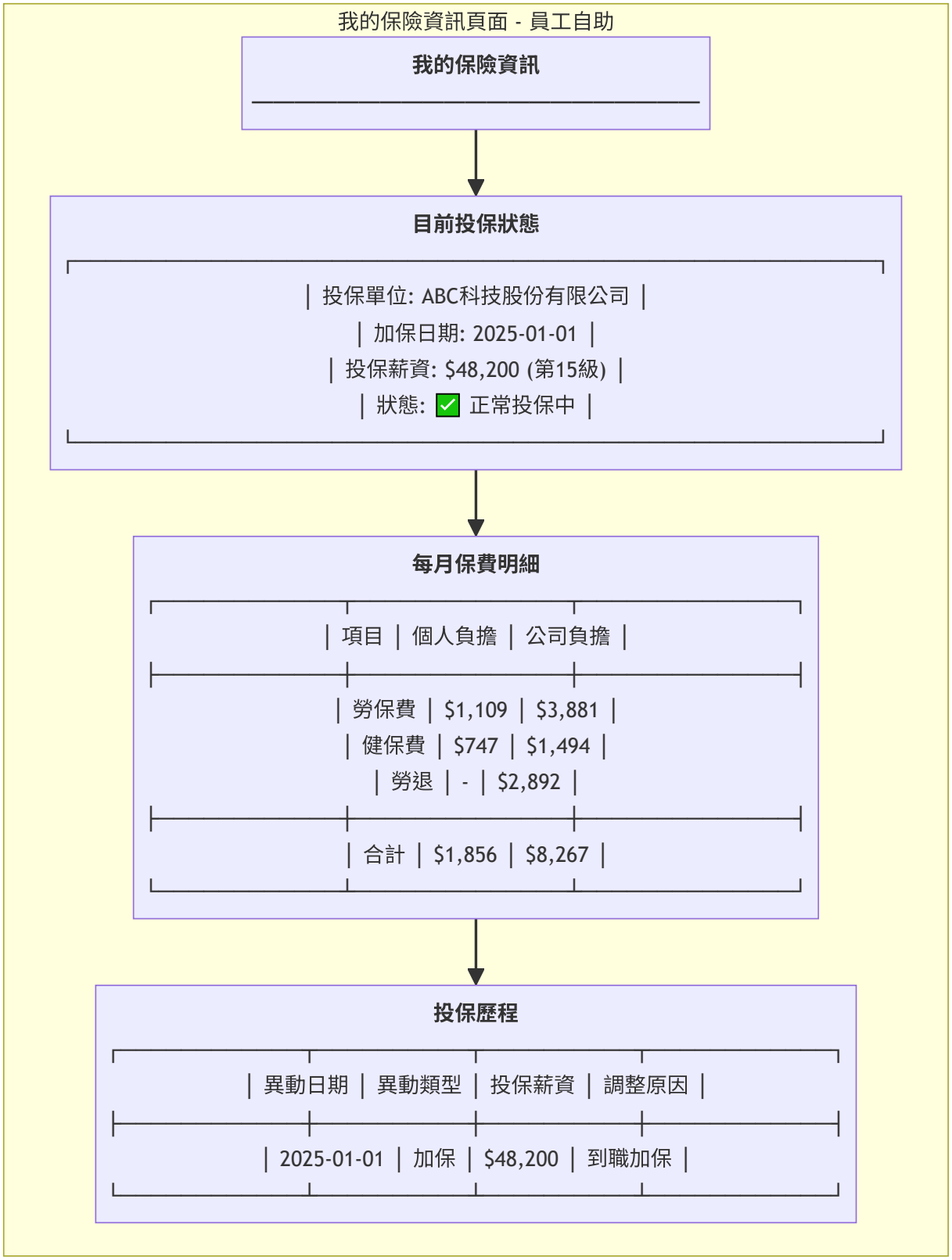
2.2.1 員工加退保記錄頁面 (HR05-P02)



2.2.2 保險費用試算頁面 (HR05-P04)

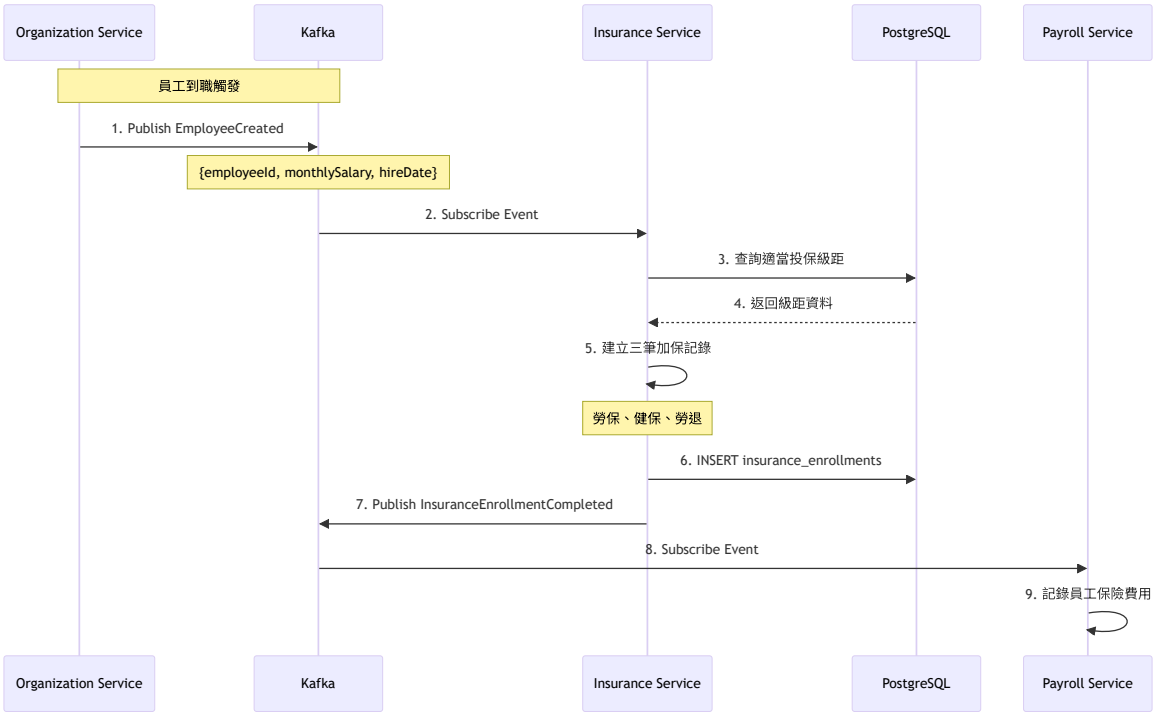
```
┌─────────────────────────────────────────────────────────────────────────────┐
│                                 保險費用試算頁面                                 │
│                                                                             │
│                                 保險費用試算                                    │
│                                                                             │
│                                 ──────────                                  │
│                                                                             │
│                                      │                                      │
│                                      ▼                                      │
│                                 輸入薪資                                        │
│                                                                             │
│                                 ┌──────────────────────────┐                  │
│                                 │ 月薪: [$_________] [🔍 試算] │                  │
│                                 └──────────────────────────┘                  │
│                                      │                                      │
│                                      ▼                                      │
│                                 試算結果 (月薪: $50,000)                        │
│                                 ┌──────────────────────────┐                  │
│                                 │ 對應投保級距: 第15級 (投保薪資: $48,200) │                  │
│                                 │                          │                  │
│                                 │ 【勞工保險】                 │                  │
│                                 │ 費率: 11.5%              │                  │
│                                 │ 員工負擔: $1,109 (20%)   │                  │
│                                 │ 雇主負擔: $3,881 (70%)   │                  │
│                                 │                          │                  │
│                                 │ 【健康保險】                 │                  │
│                                 │ 費率: 5.17%              │                  │
│                                 │ 員工負擔: $747 (30%)     │                  │
│                                 │ 雇主負擔: $1,494 (60%)   │                  │
│                                 │                          │                  │
│                                 │ 【勞工退休金】               │                  │
│                                 │ 雇主提繳: $2,892 (6%)    │                  │
│                                 │──────────────────────────│                  │
│                                 │ 員工每月負擔總計: $1,856     │                  │
│                                 │ 雇主每月負擔總計: $8,267     │                  │
│                                 └──────────────────────────┘                  │
└─────────────────────────────────────────────────────────────────────────────┘
```

2.2.3 我的保險資訊頁面 - ESS (HR05-P07)

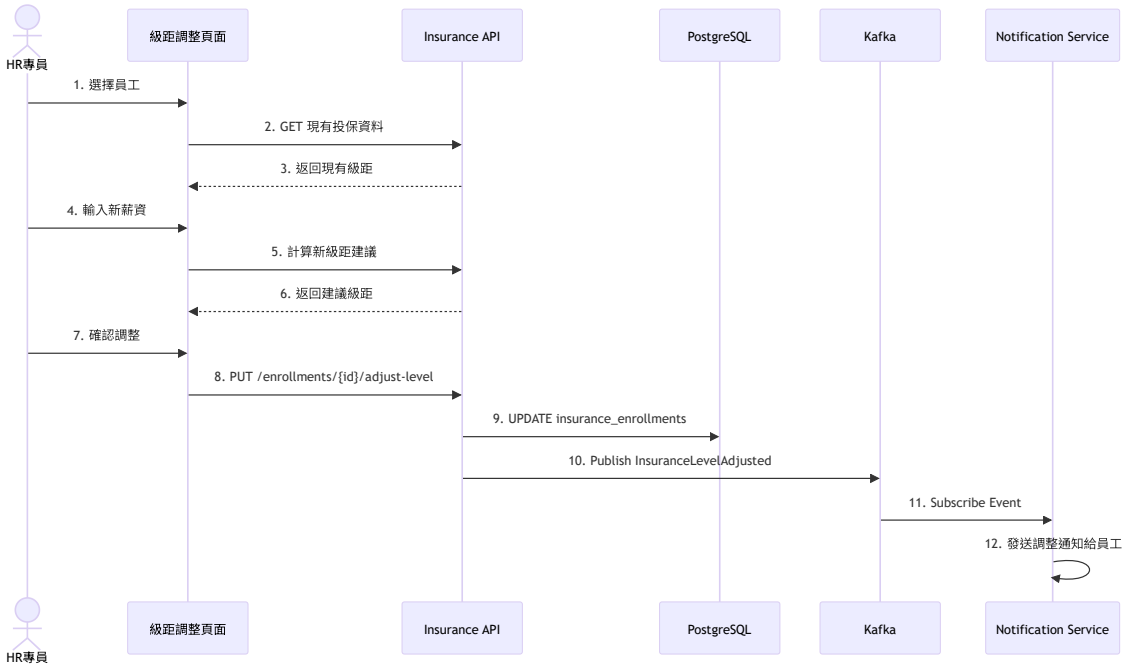


3. UX流程設計

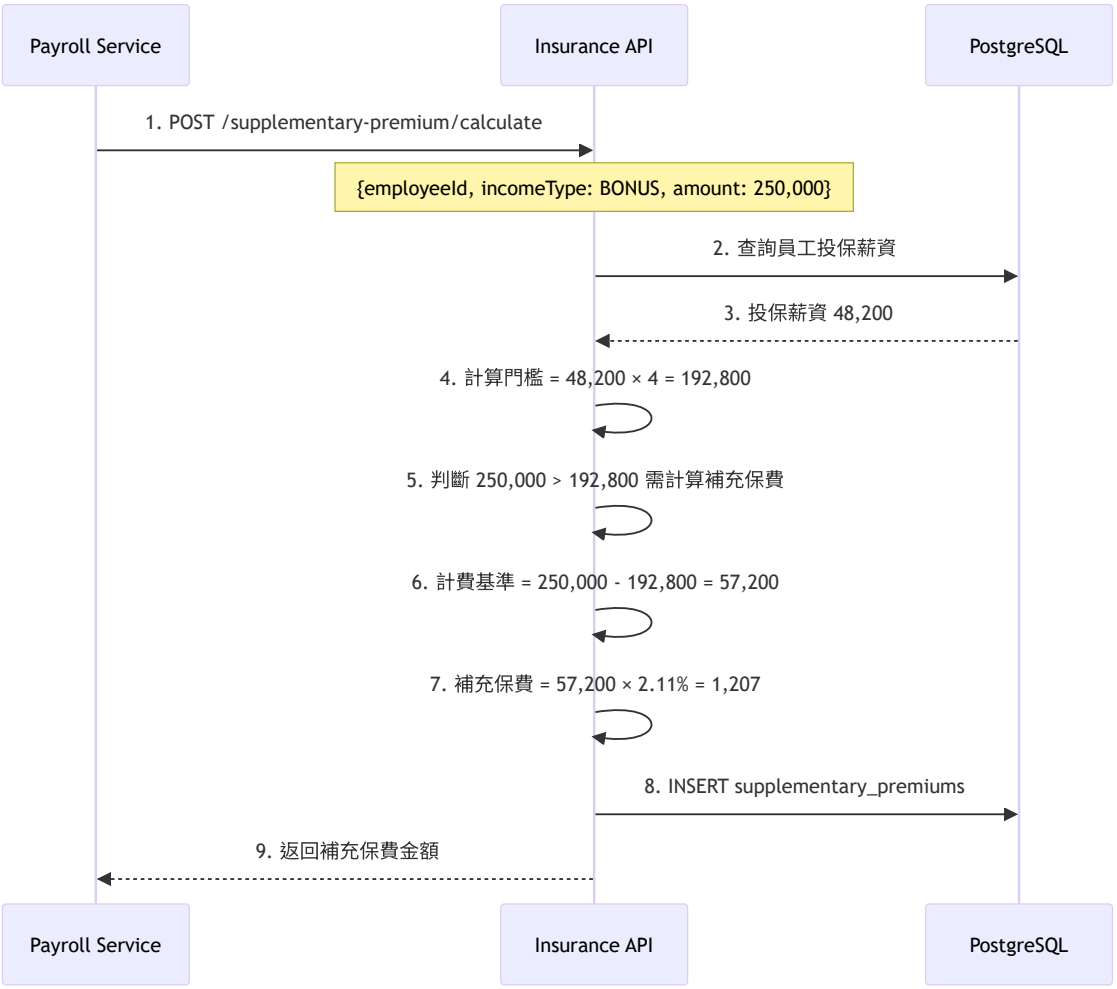
3.1 自動加保流程 (Event-Driven)



3.2 投保級距調整流程


HR專員
級距調整頁面
Insurance API
PostgreSQL
Kafka
Notification Service
1. 選擇員工
2. GET 現有投保資料
3. 返回現有級距
4. 輸入新薪資
5. 計算新級距建議
6. 返回建議級距
7. 確認調整
8. PUT /enrollments/{id}/adjust-level
9. UPDATE insurance_enrollments
10. Publish InsuranceLevelAdjusted
11. Subscribe Event
12. 發送調整通知給員工
HR專員
級距調整頁面
Insurance API
PostgreSQL
Kafka
Notification Service

3.3 補充保費計算流程



4. 畫面事件說明

4.1 加退保記錄頁面事件 (HR05-P02)

事件ID	觸發元素	事件類型	事件處理	後端API
`E-INS-01`	搜尋框	onChange	篩選員工記錄	GET /api/v1/insurance/enrollments
`E-INS-02`	手動加保按鈕	onClick	開啟加保對話框	-
`E-INS-03`	確認加保	onClick	執行加保	POST /api/v1/insurance/enrollments
`E-INS-04`	匯出申報檔	onClick	產生申報檔案	POST /api/v1/insurance/export
`E-INS-05`	調整級距按鈕	onClick	開啟級距調整對話框	-

4.2 我的保險資訊頁面事件 (HR05-P07)

事件ID	觸發元素	事件類型	事件處理	後端API
`E-MY-01`	頁面載入	onMount	載入有效投保資料	GET /api/v1/insurance/enrollments/active

4.3 費用試算頁面事件 (HR05-P04)

事件ID	觸發元素	事件類型	事件處理	後端API
`E-CALC-01`	試算按鈕	onClick	試算保費	POST /api/v1/insurance/fees/calculate

---

5. Data Flow設計

5.1 前端State結構

```
interface InsuranceState {
  units: {
    list: InsuranceUnit[];
    loading: boolean;
  };
  enrollments: {
    list: InsuranceEnrollment[];
    selectedEmployee: string | null;
    loading: boolean;
  };
  levels: {
    laborLevels: InsuranceLevel[];
    healthLevels: InsuranceLevel[];
    loading: boolean;
  };
  calculator: {
    salary: number | null;
    result: CalculationResult | null;
    loading: boolean;
  };
  myInsurance: {
    data: MyInsuranceInfo | null;
    loading: boolean;
  };
}
```

---

6. 資料庫設計

6.1 DDL Script

```
-- 投保單位表
CREATE TABLE insurance_units (
    unit_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    organization_id UUID NOT NULL,
    unit_code VARCHAR(50) NOT NULL,
    unit_name VARCHAR(255) NOT NULL,
    labor_insurance_number VARCHAR(50),
    health_insurance_number VARCHAR(50),
    pension_number VARCHAR(50),
    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT uk_unit_code UNIQUE (unit_code)
);

-- 投保級距表
CREATE TABLE insurance_levels (
    level_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    insurance_type VARCHAR(20) NOT NULL CHECK (insurance_type IN ('LABOR', 'HEALTH', 'PENSI
    level_number INTEGER NOT NULL,
    monthly_salary DECIMAL(10,2) NOT NULL,
    labor_employee_rate DECIMAL(6,4),
    labor_employer_rate DECIMAL(6,4),
    health_employee_rate DECIMAL(6,4),
    health_employer_rate DECIMAL(6,4),
    pension_employer_rate DECIMAL(6,4) DEFAULT 0.06,
    effective_date DATE NOT NULL,
    end_date DATE,
    is_active BOOLEAN DEFAULT TRUE,
    CONSTRAINT uk_level UNIQUE (insurance_type, level_number, effective_date)
);

-- 加退保記錄表
CREATE TABLE insurance_enrollments (
    enrollment_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    employee_id UUID NOT NULL,
    insurance_unit_id UUID NOT NULL REFERENCES insurance_units(unit_id),
    insurance_type VARCHAR(20) NOT NULL CHECK (insurance_type IN ('LABOR', 'HEALTH', 'PENSI
    enroll_date DATE NOT NULL,
    withdraw_date DATE,
    insurance_level_id UUID REFERENCES insurance_levels(level_id),
    monthly_salary DECIMAL(10,2) NOT NULL,
    status VARCHAR(20) NOT NULL DEFAULT 'ACTIVE' CHECK (status IN ('PENDING', 'ACTIVE', 'WI
    is_reported BOOLEAN DEFAULT FALSE,
    reported_at TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
```

```sql
    CONSTRAINT uk_enrollment UNIQUE (employee_id, insurance_type, enroll_date)
);

CREATE INDEX idx_enrollment_emp ON insurance_enrollments(employee_id);
CREATE INDEX idx_enrollment_status ON insurance_enrollments(status);

-- 補充保費記錄表
CREATE TABLE supplementary_premiums (
    premium_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    employee_id UUID NOT NULL,
    income_type VARCHAR(30) NOT NULL CHECK (income_type IN ('BONUS', 'PART_TIME_INCOME', 'P
    income_date DATE NOT NULL,
    income_amount DECIMAL(12,2) NOT NULL,
    insured_salary DECIMAL(10,2) NOT NULL,
    threshold DECIMAL(12,2) NOT NULL,
    premium_base DECIMAL(12,2) NOT NULL,
    premium_amount DECIMAL(10,2) NOT NULL,
    year INTEGER NOT NULL,
    month INTEGER NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_supp_emp ON supplementary_premiums(employee_id);
CREATE INDEX idx_supp_year ON supplementary_premiums(year, month);

-- 保險費率設定表（可調整）
CREATE TABLE insurance_rates (
    rate_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    rate_type VARCHAR(30) NOT NULL,
    rate_value DECIMAL(6,4) NOT NULL,
    effective_date DATE NOT NULL,
    end_date DATE,
    is_active BOOLEAN DEFAULT TRUE
);
```

6.2 初始化資料

```sql
-- 初始化投保級距（2025年勞保）
INSERT INTO insurance_levels (insurance_type, level_number, monthly_salary, labor_employee_
('LABOR', 1, 27470, 0.023, 0.0805, '2025-01-01'),
('LABOR', 2, 27600, 0.023, 0.0805, '2025-01-01'),
('LABOR', 3, 28800, 0.023, 0.0805, '2025-01-01'),
-- ... 更多級距
('LABOR', 15, 48200, 0.023, 0.0805, '2025-01-01'),
('LABOR', 16, 50600, 0.023, 0.0805, '2025-01-01'),
('LABOR', 17, 53000, 0.023, 0.0805, '2025-01-01'),
('LABOR', 18, 45800, 0.023, 0.0805, '2025-01-01');

-- 初始化費率
INSERT INTO insurance_rates (rate_type, rate_value, effective_date) VALUES
('LABOR_TOTAL', 0.115, '2025-01-01'),
('HEALTH_TOTAL', 0.0517, '2025-01-01'),
('PENSION_EMPLOYER', 0.06, '2025-01-01'),
('SUPPLEMENTARY', 0.0211, '2025-01-01');
```

---

7. Domain設計

7.1 聚合根

InsuranceEnrollment聚合根

```
@Entity
@Table(name = "insurance_enrollments")
public class InsuranceEnrollment {
    @EmbeddedId
    private EnrollmentId id;

    @Column(name = "employee_id", nullable = false)
    private UUID employeeId;

    @Column(name = "insurance_unit_id", nullable = false)
    private UUID insuranceUnitId;

    @Enumerated(EnumType.STRING)
    @Column(name = "insurance_type", nullable = false)
    private InsuranceType insuranceType;

    @Column(name = "enroll_date", nullable = false)
    private LocalDate enrollDate;

    @Column(name = "withdraw_date")
    private LocalDate withdrawDate;

    @Column(name = "monthly_salary", nullable = false)
    private BigDecimal monthlySalary;

    @Enumerated(EnumType.STRING)
    @Column(name = "status", nullable = false)
    private EnrollmentStatus status;

    /**
     * 建立加保記錄
     */
    public static InsuranceEnrollment enroll(
            UUID employeeId,
            UUID unitId,
            InsuranceType type,
            InsuranceLevel level,
            LocalDate enrollDate) {

        InsuranceEnrollment enrollment = new InsuranceEnrollment();
        enrollment.id = EnrollmentId.generate();
        enrollment.employeeId = employeeId;
        enrollment.insuranceUnitId = unitId;
        enrollment.insuranceType = type;
```

```java
        enrollment.enrollDate = enrollDate;
        enrollment.monthlySalary = level.getMonthlySalary();
        enrollment.status = EnrollmentStatus.ACTIVE;

        return enrollment;
    }

    /**
     * 退保
     */
    public void withdraw(LocalDate withdrawDate) {
        if (this.status != EnrollmentStatus.ACTIVE) {
            throw new DomainException("只有已加保狀態可以退保");
        }
        this.withdrawDate = withdrawDate;
        this.status = EnrollmentStatus.WITHDRAWN;
    }

    /**
     * 調整投保級距
     */
    public void adjustLevel(InsuranceLevel newLevel) {
        this.monthlySalary = newLevel.getMonthlySalary();
    }
}
```

7.2 費用計算Domain Service

```java
@Service
public class InsuranceFeeCalculationService {

    private static final BigDecimal LABOR_EMPLOYEE_RATIO = new BigDecimal("0.20");
    private static final BigDecimal LABOR_EMPLOYER_RATIO = new BigDecimal("0.70");
    private static final BigDecimal HEALTH_EMPLOYEE_RATIO = new BigDecimal("0.30");
    private static final BigDecimal HEALTH_EMPLOYER_RATIO = new BigDecimal("0.60");
    private static final BigDecimal PENSION_EMPLOYER_RATE = new BigDecimal("0.06");
    private static final BigDecimal SUPPLEMENTARY_RATE = new BigDecimal("0.0211");

    public InsuranceFees calculate(InsuranceEnrollment enrollment, InsuranceRates rates) {
        BigDecimal salary = enrollment.getMonthlySalary();

        // 勞保費用
        BigDecimal laborTotal = salary.multiply(rates.getLaborRate());
        BigDecimal laborEmployee = laborTotal.multiply(LABOR_EMPLOYEE_RATIO);
        BigDecimal laborEmployer = laborTotal.multiply(LABOR_EMPLOYER_RATIO);

        // 健保費用
        BigDecimal healthTotal = salary.multiply(rates.getHealthRate());
        BigDecimal healthEmployee = healthTotal.multiply(HEALTH_EMPLOYEE_RATIO);
        BigDecimal healthEmployer = healthTotal.multiply(HEALTH_EMPLOYER_RATIO);

        // 勞退
        BigDecimal pensionEmployer = salary.multiply(PENSION_EMPLOYER_RATE);

        return new InsuranceFees(
            laborEmployee, laborEmployer,
            healthEmployee, healthEmployer,
            pensionEmployer
        );
    }

    /**
     * 計算補充保費
     */
    public SupplementaryPremiumResult calculateSupplementary(
            UUID employeeId,
            BigDecimal incomeAmount,
            IncomeType incomeType,
            BigDecimal insuredSalary) {

        // 門檻 = 投保薪資 × 4
        BigDecimal threshold = insuredSalary.multiply(new BigDecimal("4"));

        if (incomeAmount.compareTo(threshold) <= 0) {
            return SupplementaryPremiumResult.notRequired();
        }
```

```
        // 計費基準
        BigDecimal premiumBase = incomeAmount.subtract(threshold);

        // 上限1000萬
        BigDecimal maxBase = new BigDecimal("10000000");
        if (premiumBase.compareTo(maxBase) > 0) {
            premiumBase = maxBase;
        }

        // 補充保費
        BigDecimal premiumAmount = premiumBase.multiply(SUPPLEMENTARY_RATE)
            .setScale(0, RoundingMode.HALF_UP);

        return new SupplementaryPremiumResult(
            true, threshold, premiumBase, premiumAmount
        );
    }
}
```

---

8. 領域事件設計

事件名稱	觸發時機	發布服務	訂閱服務
`InsuranceEnrollmentCompleted`	加保完成	Insurance	Payroll
`InsuranceWithdrawalCompleted`	退保完成	Insurance	Payroll
`InsuranceLevelAdjusted`	投保級距調整	Insurance	Payroll, Notification
`SupplementaryPremiumCalculated`	補充保費計算	Insurance	Payroll
`EmployeeTerminated` (訂閱)	員工離職	Organization	**Insurance**

8.2 離職連動自動退保流程

觸發： 訂閱 `EmployeeTerminated` 事件（Kafka topic: `hrms.employee.terminated`）

流程：

1. `EmployeeTerminatedEventListener` 接收事件
2. 呼叫 `AutoWithdrawOnTerminationService.withdrawAllOnTermination()`
3. 查詢員工所有 ACTIVE 加保記錄
4. 依保險類型透過 `EffectiveDateRule.calculateWithdrawDate()` 計算退保日期
5. 逐筆執行 `enrollment.withdraw(withdrawDate)` 並儲存
6. 發布 `InsuranceWithdrawalCompletedEvent`（供 Payroll 訂閱）

退保日期規則：

- 勞保/勞退/團保：離職日退保
- 健保：月底離職者次月 1 日退保（健保法規定）

8.1 事件範例

```
{
  "eventId": "evt-ins-001",
  "eventType": "InsuranceEnrollmentCompleted",
  "timestamp": "2025-01-01T09:00:00Z",
  "payload": {
    "employeeId": "emp-001",
    "enrollments": [
      {"type": "LABOR", "monthlySalary": 48200, "enrollDate": "2025-01-01"},
      {"type": "HEALTH", "monthlySalary": 48200, "enrollDate": "2025-01-01"},
      {"type": "PENSION", "monthlySalary": 48200, "enrollDate": "2025-01-01"}
    ],
    "fees": {
      "laborEmployee": 1109,
      "laborEmployer": 3881,
      "healthEmployee": 747,
      "healthEmployer": 1494,
      "pensionEmployer": 2892
    }
  }
}
```

---

9. API設計

9.1 Controller命名對照

Controller	說明
`HR05EnrollmentCmdController`	加退保Command操作
`HR05EnrollmentQryController`	加退保Query操作
`HR05FeeCmdController`	費用計算Command操作
`HR05LevelQryController`	投保級距Query操作
`HR05ExportCmdController`	申報檔案匯出

9.2 API總覽 (10個端點)

端點	方法	說明
`/api/v1/insurance/enrollments`	POST	手動加保
`/api/v1/insurance/enrollments`	GET	查詢加退保記錄
`/api/v1/insurance/enrollments/active`	GET	查詢員工有效加保記錄 (HR05-P07 頁面載入)
`/api/v1/insurance/enrollments/{id}/withdraw`	PUT	退保
`/api/v1/insurance/enrollments/{id}/adjust-level`	PUT	調整投保級距
`/api/v1/insurance/fees/calculate`	POST	計算保費 (供Payroll調用)
`/api/v1/insurance/levels`	GET	查詢投保級距
`/api/v1/insurance/supplementary-premium/calculate`	POST	計算補充保費
`/api/v1/insurance/export/enrollment-report`	POST	匯出加退保申報檔
`/api/v1/insurance/my`	GET	查詢我的保險資訊 (ESS)

10. 工項清單摘要

前端開發工項

1. HR05-P02 員工加退保記錄頁面
2. HR05-P04 保險費用試算頁面
3. HR05-P07 我的保險資訊頁面 (ESS)

後端開發工項

1. InsuranceEnrollment聚合根
2. InsuranceFeeCalculationService
3. SupplementaryPremiumCalculator
4. 加退保API (4端點)
5. 費用計算API (2端點)
6. 申報檔案產生 (勞保局/健保局格式)
7. 訂閱EmployeeCreated/EmployeeTerminated事件

8. 離職連動自動退保 (AutoWithdrawOnTerminationService + KafkaListener)

資料庫開發工項

1. 建立5個資料表DDL
2. 初始化投保級距資料
3. 初始化費率設定

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